Accenture Technical Mock Test — Set 1

45 Questions • Computer Science • Networking • Pseudo Code

Pattern aligned to the supplied Accenture mock materials. Questions are uniquely authored across the 10 sets.

Section 1 — Computer Science

Q1. Which shortest-path algorithm assumes non-negative edge weights?

A. Dijkstra's algorithm
B. Kruskal's algorithm
C. Prim's algorithm
D. Topological sort

Q2. Which SQL join forms every possible combination of rows from two tables?

A. INNER JOIN
B. LEFT JOIN
C. RIGHT JOIN
D. CROSS JOIN

Q3. Which CSS property controls the space inside an element's border?

A. gap
B. padding
C. outline
D. margin

Q4. A `Circle` class supplies its own implementation of inherited `draw()`. Which OOP feature is being used?

A. Aggregation
B. Method overloading
C. Method overriding
D. Constructor chaining

Q5. Which Java construct guarantees execution of cleanup code after try/catch?

A. throws
B. throw
C. finally
D. assert

Q6. What is the result of Java expression `"A".repeat(12).length()`?

A. 12
B. Compilation error
C. 13
D. 24

Q7. In the AWS shared responsibility model, who normally patches the guest OS of an EC2 instance?

A. ISP
B. DNS registrar
C. Cloud provider
D. Customer

Q8. Which keyword refers to the current object?

A. current
B. self
C. this
D. object

Q9. Which architecture breaks an application into independently deployable services?

A. Batch-only
B. Single-tier
C. Monolith
D. Microservices

Q10. Which cloud model provides virtual machines, networks and storage as services?

A. DBaaS only
B. SaaS
C. PaaS
D. IaaS

Q11. Which SQL operator is commonly used for wildcard pattern matching?

A. LIKE
B. IN
C. IS
D. MATCH_ROW

Q12. Which statement removes a table definition and its data?

A. DROP TABLE
B. DELETE
C. REMOVE TABLE
D. TRUNCATE ROW

Q13. In Java, `calc(int)` and `calc(long)` share a name but differ in parameter type. Which OOP feature is demonstrated?

A. Method overriding
B. Composition
C. Method overloading
D. Encapsulation

Q14. Which SQL aggregate function returns the average value of `credits`?

A. AVG()
B. COUNT()
C. MAX()
D. SUM()

Section 2 — Fundamentals of Networking

Q15. Which attack exploits malicious SQL input to manipulate a database query?

A. SQL injection
B. ARP spoofing
C. DNS tunneling
D. DDoS

Q16. Which protocol uses a three-way handshake before data transfer?

A. TCP
B. UDP
C. IP
D. ICMP

Q17. Which protocol commonly uses UDP 123?

A. RIP
B. NTP
C. SNMP
D. DNS

Q18. Which protocol is commonly used to securely manage network devices through a command line?

A. SSH
B. TFTP
C. HTTP
D. Telnet

Q19. Which default port is associated with HTTP?

A. 22
B. 3389
C. 80
D. 443

Q20. Which protocol prevents Layer-2 switching loops by blocking redundant paths?

A. ARP
B. STP
C. RIP
D. BGP

Q21. How many usable IPv4 host addresses are normally available in a /27 subnet?

A. 16
B. 30
C. 28
D. 32

Q22. Which system detects suspicious traffic but normally does not block it inline?

A. IPS
B. NAT
C. IDS
D. Proxy

Q23. Which protocol uses path-vector routing?

A. RIP
B. BGP
C. ARP
D. OSPF

Q24. Which technology securely connects a user device to a corporate network over the Internet?

A. STP
B. Site-to-site VPN
C. Remote-access VPN
D. VLAN trunk

Q25. Which protocol is used to automatically synchronize time across networked systems?

A. SNMP
B. RDP
C. NTP
D. LDAP

Q26. Which cloud networking construct logically isolates resources inside a public cloud?

A. Hypervisor cache
B. Artifact repository
C. BIOS
D. VPC/VNet

Q27. A /28 IPv4 subnet contains how many total addresses?

A. 16
B. 8

C. 32

D. 14

Q28. Which route is selected when multiple routes exist and has the longest matching prefix?

A. Random route

B. Default route always

C. Most specific route

D. Oldest route

Section 3 — Pseudo Code

Q29. What is printed by the pseudocode?

```
x ← 7
sum ← 0
FOR i ← 1 TO x
sum ← sum + i
END FOR
PRINT sum
```

A. 49

B. 29

C. 28

D. 27

Q30. What is the output?

```
A ← [5, 1, 3, 7]
sum ← 0
FOR i ← 0 TO 3
sum ← sum + A[i]
END FOR
PRINT sum
```

A. 7

B. 16

C. 1

D. 4

Q31. What value is printed?

```
n ← 1234
rev ← 0
WHILE n > 0
d ← n MOD 10
rev ← rev * 10 + d
n ← n DIV 10
END WHILE
PRINT rev
```

A. 4322

B. 1234

C. 4

D. 4321

Q32. What value is printed?

```
A ← [20, 15, 18, 9]
mx ← A[0]
FOR i ← 1 TO 3
IF A[i] > mx
mx ← A[i]
END IF
END FOR
PRINT mx
```

A. 4

B. 9

C. 62

D. 20

Q33. What is printed for the string `LEVEL`?

```
s ← "LEVEL"
r ← reverse(s)
IF s = r
PRINT "YES"
ELSE
PRINT "NO"
END IF
```

A. NO

B. YES

C. EMPTY

D. ERROR

Q34. What value is printed?

```
count ← 0
FOR i ← 1 TO 20
IF i MOD 2 = 0
count ← count + 1
END IF
END FOR
PRINT count
```

A. 11

B. 10

C. 20

D. 9

Q35. How many vowels are counted in `SECURITY`?

```
s ← "SECURITY"
count ← 0
FOR each ch IN s
IF ch is a vowel
count ← count + 1
END IF
END FOR
PRINT count
```

A. 4

B. 0

C. 8

D. 3

Q36. What position is printed for the target?

```
A ← [9, 2, 6, 5]
target ← 6
FOR i ← 0 TO 3
IF A[i] = target
PRINT i + 1
BREAK
END IF
END FOR
```

A. 4

B. 3

C. 0

D. 2

Q37. What is printed?

```
x ← 17
PRINT x MOD 5
```

A. 2

B. 5

C. 3

D. 3

Q38. What value is printed?

```
a ← 9
b ← 2
IF a > b
PRINT a
ELSE
PRINT b
END IF
```

A. 2

B. 11

C. 9

D. 7

Q39. What is printed after the assignments?

```
x ← 6
y ← 1
x ← y
y ← 9
PRINT x + y
```

A. -8

B. 15

C. 7

D. 10

Q40. Using bitwise OR, what is printed?

```
x ← 15
y ← 8
PRINT x | y
```

A. 120

B. 23

C. 7

D. 15

Q41. What is printed by the recursive function for n = 6?

```
FUNCTION F(n)
IF n <= 1
RETURN n
RETURN F(n-1) + F(n-2)
END FUNCTION
PRINT F(6)
```

A. 6

B. 9

C. 7

D. 8

Q42. What is the final printed value?

```
n ← 1209
sum ← 0
WHILE n > 0
digit ← n MOD 10
sum ← sum + digit
n ← n DIV 10
END WHILE
PRINT sum
```

A. 12

B. 9

C. 4

D. 13

Q43. What is printed?

```
n ← 12
fact ← 1
FOR i ← 1 TO n
fact ← fact * i
END FOR
PRINT fact
```

A. 479001601

B. 479001600

C. 144

D. 479001588

Q44. How many times is the inner statement executed?

```
count ← 0
FOR i ← 1 TO 4
FOR j ← 1 TO 2
count ← count + 1
END FOR
END FOR
PRINT count
```

A. 4

B. 8

C. 6

D. 7

Q45. What is the final value of sum?

```
sum ← 0
FOR i ← 1 TO 13 STEP 3
sum ← sum + i
END FOR
PRINT sum
```

A. 35

B. 5

C. 38

D. 12